Contents

Set 1

Model Question
Issued by CDC, 2077

Grade XI	**Full Marks: 75**
Subject Code: Phy 101	**Pass Marks: 27**
Subject: Physics	**Time: 3 hrs.**

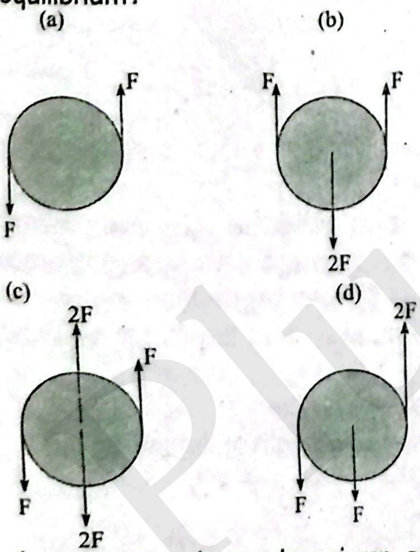
Attempt ALL questions.

Group A

Circle the best alternative to the following questions.

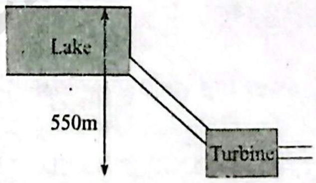
[11×1=11]

1. A metre rules used to measure the length of a piece of string in a certain experiment. It is found to be 20 cm long to the nearest millimeter. How should this result be recorded in a table of results?
   a. 0.2000 m b. 0.200 m
   c. 20 m d. 0.2 m
2. Forces are applied to a rigid body. The forces all act in the same plane. In which diagram is the body in equilibrium?

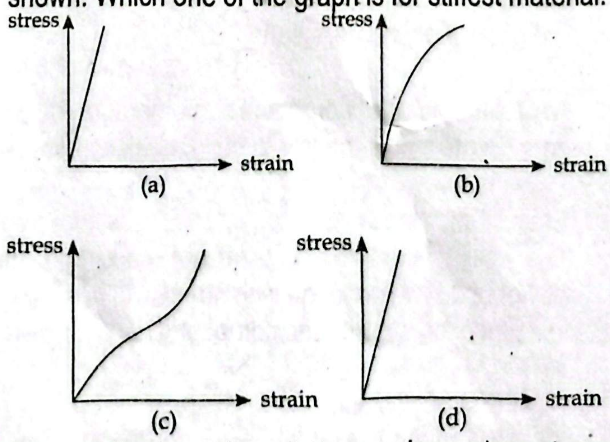


3. An athlete makes a long jump and follows a projectile motion. Air resistance is negligible. Which one of the following statement is true about the athlete?
   a. The athlete has a constant horizontal and vertical velocities.
   b. The athlete has a constant horizontal velocity and constant downward acceleration.
   c. The athlete has a constant upward acceleration followed by a constant downward acceleration.
   d. The athlete has a constant upward velocity followed by a constant downward velocity.

4. At Kulekhani-I Hydro-power station, water flows from Indra Sarowar into the turbines that are a vertical distance of 550 m below the take, as shown in the diagram. Generally, 78000 kg of water flows into the turbines every minute. The turbines have the efficiency of 85%. What is the output power of the turbines?
   a. 71 MW
   b. 60 MW
   c. 4.2 GW
   d. 3.6 GW



5. Graphs of stress-strain for four different materials are shown. Which one of the graph is for stiffest material.



6. A boy walks towards a stationary plane mirror at a speed of 1.2 ms^{-1}. What is the relative speed of approach of the boy and his image?
   a. zero b. 1.2 ms^{-1}
   c. 2.4 ms^{-1} d. 1.44 ms^{-1}

7. The critical angle between an equilateral prism and air is 45°. What happens to the incident ray perpendicular to the refracting surface?
   a. It is reflected totally from the second surface and emerges perpendicular from the third surface.
   b. It gets reflected from second and third surfaces and emerges from the first surface.
   c. It keeps reflecting from all the three sides of the prism and never emerges out.
   d. After deviation, it gets refracted from the second surface.
8. In the formation of a rainbow, the light from the sun on water droplets undergoes which of the following phenomenon/phenomena?
   a. dispersion only
   b. only total reflection
   c. dispersion and total internal reflection
   d. scattering
9. In what unit is the power of lens measured?
   a. watt b. metre
   c. dioptre d. Hertz
10. A piece of wire of resistance R is bent through 180° at mid-point and the two halves are twisted together. What is the resistance of the wire thus formed?
   a. R/4 b. R/2
   c. R d. 2R
11. What are the elementary particles with half spin called?
   a. quarks b. bosons
   c. fermions d. hadrons

Answers										
1.	2.	3.	4.	5.	6.	7.	8.	9.	10.	11.
b	b	b	b	b	c	a	c	c	a	c

Group B

Answer the following questions. **[8×5=40]**

12. a. State the law of conservation of momentum. [2]
   b. A jumbo jet of mass 4×10^2 kg travelling at a speed of 5000 m/s lands on the airport. It takes 2 minutes to come to rest. Calculate the average force applied by the ground on the aeroplane. [2]
   Ans: – 1.67×10^7 N
   c. After landing the aeroplane's momentum becomes zero. Explain how the law of conservation holds here.[2]

OR

   a. State Hook's law. [2]
   b. The walls of the tyres on a car are made of a rubber compound. The variation with stress of the strain of a specimen of this rubber compound is shown in fig. 1.2.
   As the car moves, the walls of the tyres end and straighten continuously. Use Fig. 1.2 to explain why the walls of the tyres become warm.

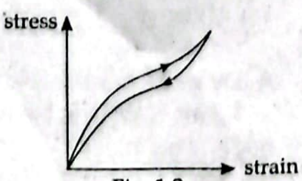


Fig. 1.2

13. a. What is meant by specific latent heat of vaporization of water=2.26 $MJkg^{-1}$? [1]
   b. A 1.0 kW kettle contains 500 g of boiling water. Calculate the time needed to evaporate all the water in the kettle. (Specific latent heat of vaporization of water = 2.26 $MJkg^{-1}$). [3]
   Ans: 1.13×10^3 sec.
   c. Explain why the actual time needed is a little longer than the time calculated in 2(b). [1]
14. a. State any three properties of an ideal gas as assumed by the kinetic theory of gas. [3]
   b. A student needed to use the ideal gas for a certain experiment. But, the ideal gas does not exist. Suggest what two different things this student could do to solve his problem. [2]
15. a. Define temperature gradient in an object. [1]
   b. An electric kitchen range has a total wall area of 1.40 m^2 and is insulted with a layer of fiber glass that has a temperature of 175°C and its outside surface is 35°C. The fiber glass of 4.0 cm thick has a thermal conductivity of 0.040 $Wm^{-1}K^{-1}$. Calculate the rate of flow of heat through the insulation, assuming the fibre as a flat slab of area of 1.40 m^2. [3]
   Ans: 210 W
   c. How might the rate of conduction be affected if the fiber absorbs moisture? Justify your answer. [1]
16. Figure 5.1 shows a ray of light is entering and emerging through a part of a convex lens.
   a. Define 'convex lens' and state one daily application of it. [2]
   b. Explain why this lens is also called converging lens?[1]
   c. Calculate the refractive index of the material of the lens shown in the figure.

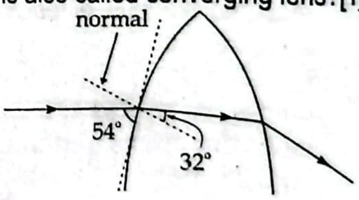


Fig. 5.1

   Ans: (c) 1.5

OR

   a. Define 'concave mirror' and state one daily application of it. [2]
   b. A certain projector uses a concave mirror for projecting an object's image on a screen. It produces an image that is 5 times bigger than the object and the recent is 5 m away from the mirror as shown in Fig. 5.2.
      i. Given reason why is the
      ii. Calculate the focal length of the mirror. [2]

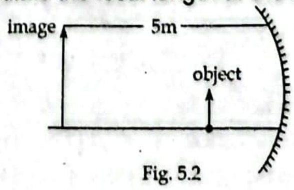


Fig. 5.2

   Ans: (ii) 0.53 m

17. a. Sketch an electric field pattern around two identical negative point charges shown below.

b. Obtain an equation, in terms of Q and r, for the field strength at point X due to two charges shown in shown in Fig. 6.1. [3]

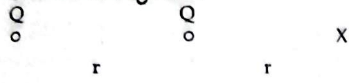


Fig. 6.1

18. a. Define capacitance of a parallel plate capacitor and state one application of it in electric circuit. [2]

b. Three capacitors each of 1000 μF are connected in an electric circuit as shown below.

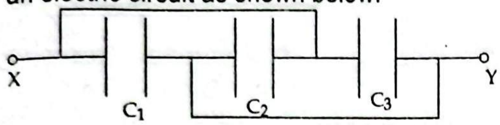


Fig. 7.1

i. Identify the type of combination shown in fig. 7.1 and calculate the effective capacitance of the combination. [1+2]

Ans: 3000 μF

19. a. What is it meant by power of a heater is 2 kW? [1]

b. Calculate the resistance of the above mentioned heater when it is connected to 220V source. [2]

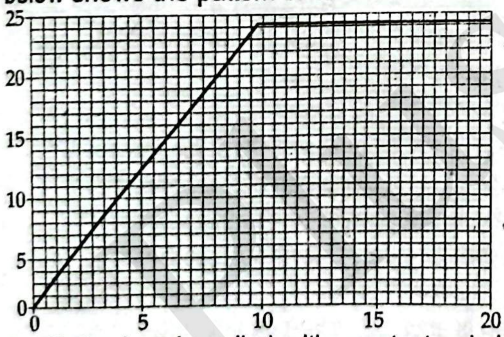
Ans: 24.2 Ω

c. Suggest what changes must be done to the heater so that it gives more heat. Justify your answer. [2]

Group C

Give long answer in the following questions. [3×8=24]

20. A box rest is accelerated by a rope attached with a motor as shown in the fig. 2.1. The velocity-time graph given below shows the pattern of its motion for 20s.



a. If the box is pulled with constant unbalanced force 10N. Show that the initial acceleration of the box is 2.5 ms^{-1}, and calculate its mass. [2+1]

Ans: 4 kg

b. After 2.0 second the box is being pulled by a constant force 12N. Determine the size of frictional forces acting on the box at this time. [2]

Ans: 2 N

c. Determine the distance of the box travels along the ground at 8.0s. [3]

Ans: 80 m

21. A boy is operating a remote controlled toy car on a horizontal-circular track, as shown in figure. The track has a radius of 1.8 m and the car travels around the track with a constant speed.

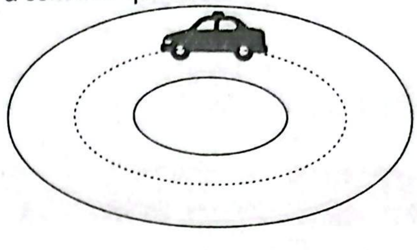


Fig. 10.1

i. Explain why the car is accelerating even though it is travelling at a constant speed. [2]

ii. The car has a mass of 0.50 kg. The boy now increases the speed of the car to 6-0 ms^{-1}.
The total radial friction between the car and the track has a maximum value of 7.0 N. Show by calculation that the car cannot continue to travel in a circular path. [3]

iii. The car is now placed on a track, which includes a raised section. This is shown in Figure 10.2.

The raised section of the truck can be considered as the arc of a circle, which has radius of 0.85 m. The car will lose contact with the raised section of track if its speed is greater than v_{max}. Show that v_{max} is given by the relationship v_{max}. [3]

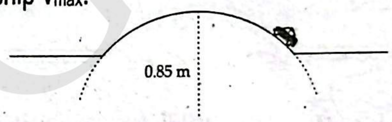
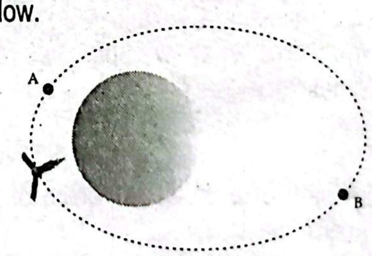


Fig. 10.2

OR

Juno is a NASA orbiter with a mission to survey Jupiter. It is in an elliptical orbit around Jupiter as shown in the figure below.



The gravitational potential at point A in the orbit of Juno is- 1.70×10^9 Jkg^{-1}.

a. State what is meant by a gravitational potential at point A is -1.70×10^9 Jkg^{-1}. [2]

b. At point B Juno is 1.69×10^8 m from the centre of Jupiter. If the mass if Jupiter is 1.90×10^{27} kg, calculate the gravitational potential at point B. [3]

c. The mass of Juno is 1.6×10^7 kg. Determine the change in gravitation potential energy if Juno moves from point A to point B. [3]

Ans: (b) 7.5×10^8 J Kg^{-1} (c) 1.52×10^{16} J

22. a. Explain how Rutherford's α-scattering experiment suggested that the nucleus of an atom is very small, very dense and positively charged. [3]

b. Considering that the α-particles carry average kinetic energy of 2.00×10^{-10} J. Calculate the maximum size of the gold nucleus. (Atomic number of gold is 79 and $e = 1.60 \times 10^{-19}$ C) [3]

Ans: 3.6×10^{-16} m

c. Explain why the radius of the gold nucleus must be much smaller than the value calculated in 11 (b) above. [2]

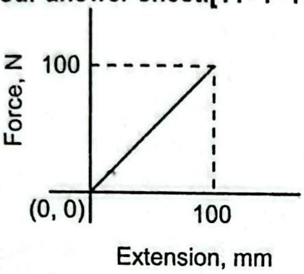
Set 2: HISSAN 2082 (2025)

GROUP A

Multiple Choice Questions.

Rewrite the correct option in your answer sheet.[11×1=11]

1. A force versus extension graph of a spring is shown in the figure. What is the work done in extending the spring?



a. 10J b. 5J
c. 5000J d. 500J

2. Newton's law of gravitation applies to
a. small bodies only b. planets only
c. larger bodies only d. all bodies irrespective of their size

3. If two non-zero vectors $\vec{A}$ and $\vec{B}$ obey the relation $|\vec{A} + \vec{B}| = |\vec{A} - \vec{B}|$, what will be the angle between $\vec{A}$ and $\vec{B}$?
a. 0° b. 45°
c. 90° d. 180°

4. Two bodies of masses 4 kg and 5 kg are moving with equal momentum. What will be the ratio of their respective kinetic energies?
a. 2 : 1 b. 1 : 3
c. 5 : 4 d. 4 : 5

5. In the entire path of a projectile, which of the following quantity remains constant?
a. vertical component of velocity
b. horizontal component of velocity
c. kinetic energy
d. potential energy

6. The total internal reflection takes place when light passes from
a. air to water b. air to glass
c. water to glass d. glass to water

7. Which mirror is preferred for shaving and make up purposes?
a. Plane mirror b. Convex mirror
c. Concave mirror d. Both convex and plane mirror

8. What will be the velocity of the light ray in a medium which has refractive index 1.5?
a. 2×10^8 m/s b. 1.5×10^8 m/s
c. 0.75×10^8 m/s d. 3×10^8 m/s

9. The refractive index of a medium is greatest for
a. red light b. green light
c. yellow light d. violet light

10. The number of electrons in one coulomb charge is
a. 1.6×10^{19} b. 6×10^{19}
c. 1.6×10^{18} d. 6.24×10^{18}

11. Holes are majority charge carrier in
a. N-type semiconductor b. ionic solids
c. P-type semiconductor d. non-metals

Answers

1.	2.	3.	4.	5.	6.	7.	8.	9.	10.	11.
b	d	c	c	b	d	c	a	d	d	c

GROUP B

Short Answer questions: **[5×8=40]**

12. a. Define coefficient of friction. Prove that the coefficient of limiting friction is equal to the tangent of angle of friction. [1+2]

b. A ball X of mass 0.1 kg, moving with a velocity of $6ms^{-1}$ collides directly with another ball Y of mass 0.2 kg at rest. If X had rebounded with a velocity of $2ms^{-1}$ in the opposite direction after collision, what would be the velocity of Y? [2]

Ans: 4 m/sec

OR

a. What is elastic collision? Show that in one dimensional elastic collision, the relative velocity of approach before collision is equal to the relative velocity of recession after collision. [1+2]

b. A body moving horizontally with a kinetic energy of 800 joule experiences a constant opposing horizontal force of 100 N while moving from point A to point B where AB = 2 m. What is the kinetic energy of the body at B? [2]

Ans: 600 J

13. a. Define coefficient of linear expansion of a solid. Does it depend upon the length of the conductor? [1+1]

b. What is triple point? Explain the method to determine the specific heat capacity of solid by the method of mixture. [1+2]

14. a. What is thermal equilibrium? How is it applied in working of mercury thermometer? Explain. [1+1]

b. Define emissivity. Calculate the energy radiated per minutes from the filament of a lamp at temperature 5000 K. Take the filament has surface area 4×10^{-5} m² with its relative emittance 0.85 and Stefan's constant = $5.67 \times 10^{-8} W m^{-2} K^{-4}$. [1+2]

Ans: 120 Watt

15. a. Define one mole of a gas. Why is hydrogen gas not found in the atmosphere? Explain. [1+2]

b. Assuming the density of nitrogen at STP to be 1.251 kg/m³, find the root mean square velocity of nitrogen molecules at 127°C. [2]

Ans: 597 m/sec

16. a. What is lateral shift? Depict the variation of lateral shift with the angle of incidence. Explain why there is no dispersion when light passes through a glass slab. [3]

b. A convex lens of focal length 24 cm is totally immersed in water of refractive index 1.33. Find the focal length of the lens in water. [Refractive index of glass = 1.5] [2]

Ans: 93.8 cm

OR

a. What is chromatic aberration in lens? Prove mathematically that the chromatic aberration is equal to the product of dispersive power and focal length of the mean light. [1+2]

b. When an object is placed at a distance of 40 cm in front of a concave mirror, the size of image becomes equal to the size of the object. Calculate the focal length of the mirror. [2]

Ans: 20 cm

17. a. Define electric field intensity. [1]

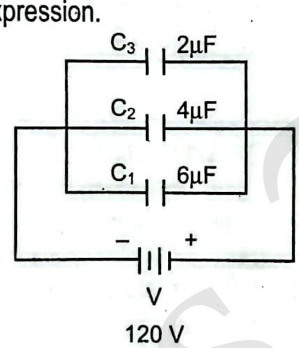
b. Find an expression of electric field intensity due to an infinite plane charge conductor. [2]

c. What is potential gradient? Give its relation in terms of electric field intensity. [1+1]

18. a. Define capacitance of a capacitor. [1]

b. Describe the theory of energy stored in a capacitor with mathematical expression. [2]

c. Calculate the equivalent capacitance and charge on each capacitor of given circuit diagram. [2]



Ans: 12 μF, 240 μC, 480 μC & 720 μC

19. a. What is drift velocity of electrons? Two cupper wires of different diameter are joined end to end. If current flows in the wire combination what happens to the drift speed of electrons when they move from larger diameter wire to smaller diameter wire. [1+2]

b. A 480 Watt electric heater is designed to operate from 240 V electric line.

i. Find the resistance of the heater. [1]

ii. Find the current through the heater. [1]

Ans: (i) 120 Ω (ii) 2A

GROUP C

Long answer questions: **[3×8=24]**

20. a. Resolve a vector into two rectangular components. [2]

b. A baseball is thrown towards a player with an initial speed of 20 m/s and at angle 45° with the horizontal. At the moment the ball is thrown, the player is 50 m far from the thrower. At what speed and in what direction must the player run to catch the ball at the same height at which it was released? [3]

Ans: 3.5 m/s, towards the thrower

c. What is elastic limit? Deduce an expression for the work done in stretching a wire. [1+2]

21. a. What is acceleration due to gravity? Obtain a formula for the variation of acceleration with altitude from the surface of the earth. [1+2]

b. A student wants to escape a body of mass 1000kg from earth's surface. Calculate the velocity required to escape the body from earth's surface. [$G = 6.67 \times 10^{-11}$ Nm^2/kg^2, $Me = 5.98 \times 10^{24}$ kg, $Re = 6.4 \times 10^6$m] [3]

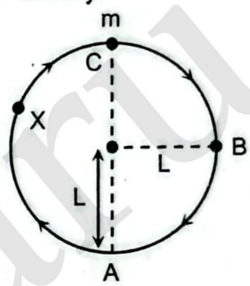
Ans: 11.18 m/sec

c. What is Global Positioning System [GPS]? Give applications of global positioning system. [2]

OR

a. Define angular velocity. Establish a relation between angular velocity and linear velocity. [1+2]

b. A bob of mass m is tied by a light string of length L and is whirled into a vertical circle with uniform velocity v as shown in figure.



i. Define centripetal force. Mention its direction. [1+1]

ii. What is tension in the string at point A and B? [1+1]

iii. If the string is cut at C, what will be the trajectory of the particle? [1]

22. a. All the nuclei have nearly the same density, why? [1]

b. Define mass defect and binding energy per nucleon.[2]

c. Obtain the expression of critical density of universe.[2]

d. What is doping? Describe the formation of N-type semiconductor. [1+2]

Set 3: HISSAN 2081 (2024) Set A

GROUP A

Multiple Choice Question

Attempts all questions. **[11 × 1 = 11]**

Rewrite the correct option in your answer sheet.

1. A dimensionless physical quantity

a. never has unit b. may have unit

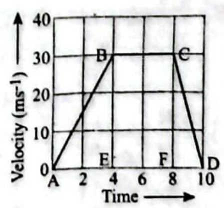
c. must have unit d. never exists

2. Dot product of two vector is equal to $\frac{1}{\sqrt{3}}$ times the magnitude of cross product. The angle between them is

a. 90° b. 30°

c. 50° d. 60°

2. The graph shows the change in velocity of an object with time in second, what is the acceleration produced along the path AB?



a. 30 m/s^2 b. 6.5 m/s^2
c. 8.0 m/s^2 d. 7.5 m/s^2

3. If you throw a ball upward with velocity of 4m/s. At what height its kinetic energy reduces to half of its initial value?
a. 4m b. 0.2m
c. 1m d. 0.4m

4. In case of elastic body if stress is increased, Young's modules
a. Decreases b. Increases
c. Remains same d, First decreases, then increases

5. An object of mass 200 g whirled round in a vertical circle of radius 2 m with a constant speed of 4 m/s. The maximum tension in the string is
a. 5.5 N b. 3.6 N
c. 4.0 N d. 5.0 N

6. Which mirror is used in the search light?
a. Plane mirror b. Convex mirror
c. Parabolic mirror d. Concave mirror

7. For a certain prism the minimum deviation angle 30°. If the angle of incidence on first refracting face is 45°, what is the refracting angle of prism?
a. 45° b. 60°
c. 75° d. 90°

8. Two lenses of focal lengths 20 cm and –40 cm are held in contact. The image of an object at infinity will be formed by the combination at
a. 40 cm b. 10 cm
c. 50 cm d. Infinite

9. A ray of light passes from vacuum into a medium of refractive index μ, the angle of incidence is found to be twice the angle of refraction. The angle of incidence is
a. $\cos^{-1}\left(\frac{\mu}{2}\right)$ b. $2\cos^{-1}\left(\frac{\mu}{2}\right)$
c. $2\sin^{-1}\left(\frac{\mu}{2}\right)$ d. $2\sin^{-1}(\mu)$

10. A piece of wire of resistance R is bent to form a circle. What is the resistance of the circular wire along its diameter?
a. R b. R/2
c. R/4 d. 2R

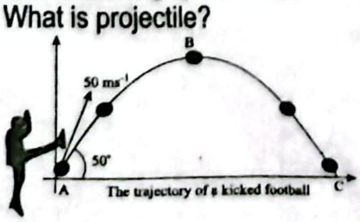
11. Which one is the particle - antiparticle pair?
a. Electron and proton b. Neutron and electron
c. Proton and neutron d. Electron and positron

Answers											
1.	2.	2.	3.	4.	5.	6.	7.	8.	9.	10.	11.
a	d	d	d	c	b	d	b	a	b	c	d

GROUP B

Answer the following questions. **[8 × 5 = 40]**

12. a. What is projectile? [1]



The trajectory of a kicked football

b. Prove that its path is parabolic. [2]
c. A ball is kicked from ground level, from point A, and strikes the ground at point C as shown in the figure.
i. Find the distance between A & C.
ii. Find the time taken by the ball to reach the point B. [2]

13. a. What is water equivalent of substance? Is it equal to heat capacity? [2]
b. Use method of mixture to determine the latent heat of vaporization with suitable diagram. [3]

14. a. What is thermal conductivity? [1]
b. Wood has low value of thermal conductivity. What does it mean? [1]
c. A metal cylinder containing water at 60°C has thickness of 4 mm and thermal conductivity equal to 400W/mK. It is lagged by felt of thickness 2mm and thermal conductivity of 0.002 W/mK. The room temperature is 10°C. Find the temperature of metal felt interface. [3]

Ans: 59.9 °C

15. a. What is differential expansion of solid? [1]
b. Develop a relation between α and ϒ, where symbols have usual meanings. [2]
c. The density of organic liquid at 0°C is 900 kg/m^3 and the coefficient of linear expansion is 0.0004/°C. Find its density at 100°C. [2]

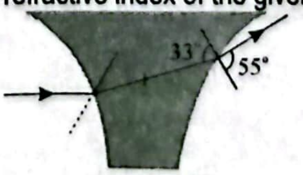
Ans: 865.38 kg/m^3

OR

a. What is ideal gas? [1]
b. Using kinetic molecular model of gas develop a relation between kinetic energy and temperature of gas. [2]
c. A helium storage tank has a capacity of 0.05 m^3. If the gas pressure is 100 atmosphere at 27°C, Determine (a) the number of moles of helium and (b) The mass of helium [R = 8.314 J $mol^{-1}K^{-1}$, relative molecular mass of helium is 4] [2]

Ans: 203.5, 0.81 kg

16. a. What is Lateral shift? Show that it is equal to thickness of glass slab when angle of incidence is 90°. [2]
b. Define concave lens and mention one of its uses in our daily life. [1]
c. The figure shows a ray of light entering and emerging through a part of concave lens. Calculate the refractive index of the given lens. [2]



Ans: 1.5

OR

a. What happens to the focal length of lens if it is dipped in water? [1]

b. Discuss the formula for the focal length of two thin lenses in contact. [2]

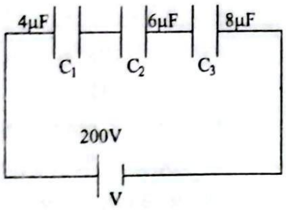
c. For a 60° glass prism, the angle of minimum deviation is 37.2°. Calculate its refractive index. [2]

Ans: 1.5

17. a. What is a capacitor? [1]

b. What are the factors affecting capacitance of a capacitor? [1]

c. In the given circuit calculate the equivalent capacitance and charge on the capacitors. [3]

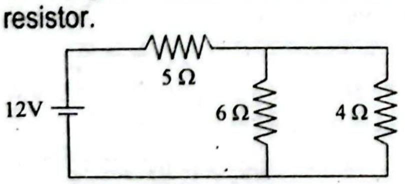


Ans: 1.85 μF, 3.69×10^{-4}C

18. a. What is drift velocity of an electron? [1]

b. Derive a relation between the current through a metallic conductor and drift velocity of electron. [2]

c. In the given circuit find the heat developed in 5Ω resistor. [2]



Ans: 28.5 J/ sec

19. a. State and explain Gauss law in electrostatics. [2]

b. What is electric potential? [1]

c. Two charges of +1 and +4 micro-coulomb respectively are placed 12cm apart. Find the position of the point where electric field intensity is zero. [2]

Ans: 0.04 m from + 1 mC

GROUP C

Answer the following questions. **[3 × 8 = 24]**

20. a. Differentiate between center of gravity and center of mass. [2]

b. Find the expression for the escape velocity of an object at the surface of the earth. [3]

c. An artificial satellite of the earth revolves in a circular orbit at a height 35900 km above the earth surface. What is the orbital speed and period of revolution of the satellite?
[$G = 6.67 \times 10^{-11}$ Nm²/kg⁻², $R = 6.4 \times 10^6$ m, $M_e = 6 \times 10^{24}$ kg] [3]

Ans: 3111.8 m/sec, 23.7 hrs

OR

a. What is impulse? Give its relation with linear momentum. [2]

b. Show that angle of friction is equal to angle of repose if the surfaces have same nature? [3]

c. A 550kw power engine of a vehicle of mass 1.2×10^5 kg is rising on an incline plane of inclination 1 in 80 with constant speed of 50 km/hr. Find the frictional force between the wheels of the vehicle and the incline plane. [3]

Ans: 40 N

21. a. How can you find the work done if force acting in the body is variable. Give idea only. [1]

b. Differentiate between elastic and inelastic collision. [2]

c. In case of stretched wire prove that: Energy density = $\frac{1}{2}$ Stress × Strain. [3]

d. A metallic wire of diameter 1mm and length 1.2m is stretched by 1/5th of 1% of its initial length by a force of 40 N. Find the value of Young's modulus of the wire. [2]

Ans: 2.55×10^{10} N/m²

22. a. Write down famous mass – energy relation and explain it's significances. [2]

b. Define the terms doping. Explain how P-type of semiconductor is formed. [2]

c. State and explain Hubble law. [2]

d. Using the values given below, calculate the binding energy for $^{238}_{92}U$
[Mass of $^{238}_{92}U$ = 238.0508 amu, m_p = 1.008665 amu, m_n = 1.008665 amu, 1amu = 931 MeV] [2]

Ans: 1782.9 MeV, 7.59 MeV/ nucleon

Set 4: HISSAN 2081 (2024) Set B

GROUP A

Multiple Choice Question

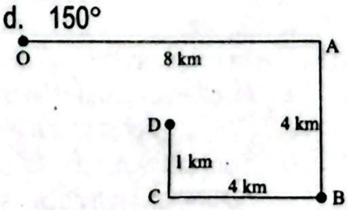
Attempts all questions. **[11 × 1 = 11]**

Rewrite the correct option in your answer sheet.

1. Density of substance is given by $\rho = at + bt^2$ where t is time, then the dimensional formula of a is
   a. [MLT^{-3}] b. [MLT^{-2}]
   c. [MLT^{-1}] d. [$ML^{-3}T^{-1}$]

2. Two forces, each of magnitude $\vec{F}$ have their resultant of same magnitude $\vec{F}$, then the angle between two forces is
   a. 120° b. 45°
   c. 60° d. 150°

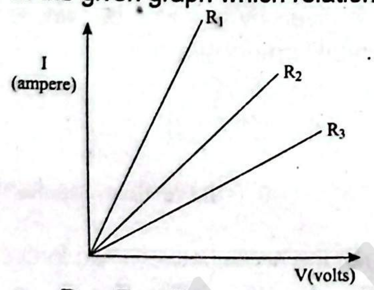
3. A car moves from O to D along the path OABCD. What is the magnitude of displacement of the car from the starting point?



   a. 3 km b. 5 km
   c. 7 km d. 17 km

4. The value of acceleration due to gravity at sea level is 9.8 m/s². Height of Manakamana temple is 1300 m from sea level. What is the acceleration due to gravity at the temple?
   a. 9.77 m/s² b. 10.0 m/s²
   c. 9.79 m/s² d. 9.80 m/s²

5. The energy stored per unit volume of stretched wire is ...
   a. $\frac{1}{2}$ load $\times$ extension b. load $\times$ stress
   c. stress $\times$ strain d. $\frac{1}{2}$ stress $\times$ strain
6. Which of the following best describes the image formed by a convex mirror?
   a. Virtual, erect and magnified
   b. Virtual, erect and diminished
   c. Real, inverted and magnified
   d. Virtual, inverted and diminished
7. The total internal reflection takes place, when light travels from -
   a. air to glass b. air to water
   c. water to glass d. glass to water
8. A convex lens of focal length f produces a real image m times the size of the object. What is the distance of object from the lens?
   a. $(m + 1)f$ b. $\left(\frac{m-1}{m}\right)f$
   c. $(m-1)f$ d. $\left(\frac{m+1}{m}\right)f$
9. An achromatism consists of two lenses whose dispersive power is in ratio 1:3. If focal length of converging lens is 30 cm, focal length of diverging lens will be
   a. 20 cm b. 40 cm
   c. 90 cm d. 120 cm
10. In the given graph which relation is correct?



   a. $R_1 > R_2 > R_3$ b. $R_1 < R_2 < R_3$
   c. $R_1 = R_2 = R_3$ d. $R_1 = R_2 > R_3$
11. The mass number of an atom is
   a. Number of protons in a nucleus
   b. The reciprocal of the atomic number
   c. Sum of the number of protons and the number of neutrons in the nucleus
   d. Number of neutrons in a nucleus

Answers										
1.	2.	3.	4.	5.	6.	7.	8.	9.	10.	11.
d	a	b	c	d	b	d	d	c	b	c

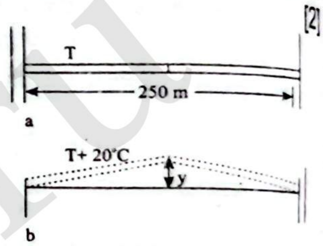
GROUP B

Answer the following questions. **[8 × 5 = 40]**

12. a. What is a projectile? [1]
   b. A ball is fired from ground making an angle θ with horizontal with a velocity v, show that path of the ball is parabolic. [2]
   c. Obtain expressions of time of flight and maximum vertical height. [2]
13. a. A person feels cold after coming out from swimming pool. Why? [2]
   b. Use the method of mixture to determine specific heat capacity of solid with suitable labelled diagram. [3]
14. a. State and explain Stefan-Boltzmann's law of black body radiation. [2]
   b. An ice box is made of wood 1.75 cm thick lined with cork 2 cm thick. The temperature of inner surface of the cork is steady at 0°C and that of outer surface of the wood is steady at 12°C. The thermal conductivity of wood is five times more than of cork. Find the temperature of interface. [3]

Ans: 10.2 °C

15. a. Define coefficient of linear expansion. Does it depend on length? Explain. [2]
   b. Two concrete spans of a 250 m long bridge are placed end to end so that no room is allowed for expansion as shown in figure (a).



   If the temperature increases by 20°C, what is the height y to which the spans rise as shown in figure b? ($\propto = 11.66 \times 10^{-6}$ K^{-1}) [3]

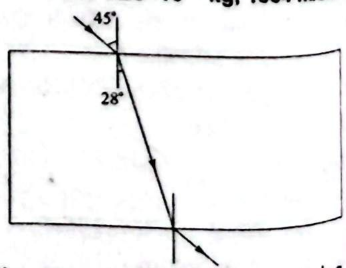
Ans: 0.42 m

OR

   a. A student explains kinetic energy of ideal gas is function of temperature. How can you show mathematically that kinetic energy of ideal gas is function of temperature? [2]
   b. Helium gas occupies a volume of 0.04 m^3 at a pressure of 2×10^5 N/m^2 and temperature 300 K. Calculate the mass of the helium and root mean square speed of its molecules.
   [relative molecular mass of helium = 4, molar gas constant = 8.3 J/mol K] [3]

Ans: 1.29×10^{-2} kg, 1364 m/sec

16. Figure shows a ray of light passing through a rectangular slab of a transparent medium.



   a. Define refractive index of the medium. [1]
   b. Mention the effect on the speed of light after entering into the medium. [1]
   c. Calculate the value of critical angle for the given medium. [3]

OR

   a. What happens to the focal length of lens if it is dipped in water? [1]

b. Write down the lens maker's formula and name the quantities associated in the formula. [2]

c. For a 60° glass prism, the angle of minimum deviation is 37.2°. Calculate its refractive index. [2]

Ans: 1.5

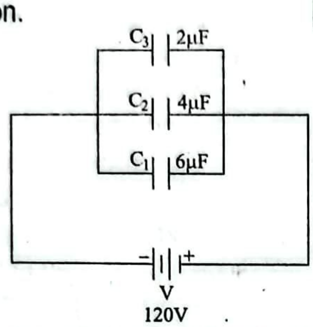
17. a. State Gauss's theorem. [1]

b. A physics professor tells that one can stay safely inside a charged sphere without electric shock. How it is possible? [1]

c. Use Gauss theorem to obtain an expression for electric field of a linearly charged body. [3]

18. a. Discuss a theory of energy stored in a capacitor with mathematical expression. [2]

b. Calculate the equivalent capacitance and charge on each capacitor in the given diagram. [3]



Ans: 12 μF, 0.24 C, 0.48 C, 0.72 C

19. a. Discuss the mechanism of metallic conduction and find the expression of drift velocity. [3]

b. What distance must an electron move in a uniform potential gradient 200 V/cm in order to gain K.E. of 3.2×10^{-18} J? Given that e = 1.6×10^{-19} C, $m_e = 9.1 \times 10^{-31}$ kg. [2]

Ans: 10^{-5} m

GROUP C

Give long answer to the following questions. [3× 8 = 24]

20. a. What are the variations of acceleration due to gravity? [1]

b. Find the expression for gravitational potential energy of a body at a point. [3]

c. What is the significance of negative value of gravitational potential energy? [1]

d. A satellite of the earth revolves in a circular orbit at a height 25000 km above the earth surface. What is the orbital speed and period of revolution of the satellite? [G = 6.67×10^{-11} Nm²/kg^{-2}, R = 6.4×10^{6} m, $M_e = 6 \times 10^{24}$ kg] [3]

Ans: 3600 m/sec, 15.2 hrs

OR

a. What is centripetal acceleration? What is its value in terms of speed and radius? [2]

b. Show that, the time period of oscillation of conical pendulum is given by $t = 2\pi\sqrt{\frac{l\cos\theta}{g}}$ where symbols have their usual meaning. [3]

c. A mass of 500 g is rotated by a string at constant speed in a vertical circle of radius 1m. If the minimum tension in the string is 3N, calculate the magnitude of speed and maximum tension in the string. [3]

Ans: 4 m/s, 13 N

21. a. State work energy theorem. [1]

b. Show that total mechanical energy of a body is conserved when it moves under the action of gravitational field. [3]

c. A bullet fired from a gun hurt more than an air molecule hitting a man, though they have almost same velocity. Why? [1]

d. A 450 N physics student stands on a bath room scale in an elevator. As the elevator starts moving the scale reads 350 N. Draw free body diagram of the problem and find the magnitude and direction for the acceleration of the elevator. [3]

22. a. Write one difference between intrinsic and extrinsic semiconductors. [1]

b. What are N-type and P-type of semiconductors? [2]

c. Define the term binding energy and draw the graph between binding energy per nucleon and mass number.[2]

d. A nucleus of $_{92}U^{238}$ disintegrates according to

$$_{92}U^{238} \longrightarrow {}_{90}Th^{234} + {}_{2}He^{4}$$

Calculate the energy released in the disintegration process if

Mass of $_{92}U^{238} = 3.859 \times 10^{-25}$ kg,

Mass of $_{90}Th^{234} = 3.787 \times 10^{-25}$ kg,

Mass of $_{2}He^{4} = 6.648 \times 10^{-27}$ kg, C = 3×10^{8} m/s. [3]

Ans: 4.236 MeV

Set 5: HISSAN 2080 (2023)

Group A

Multiple Choice Question

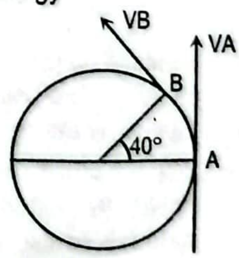
Attempts all questions. [11×1=11]

Rewrite the correct option in your answer sheet.

1. The dimensional formula for torque is
   a. $[MLT^{-1}]$ b. $[ML^2T^{-2}]$
   c. $[M^2LT^{-1}]$ d. $[ML^2T^{-1}]$

2. Newton's Second law of motion gives the measurement of
   a. force b. acceleration
   c. momentum d. energy

3. A particle is moving in a circle of radius 'r 'centered at O with a constant speed v. The change in velocity in moving from A to B is



   a. 2v cos 40° b. 2v sin 40°
   c. 2v cos 20° d. 2v sin 20°

4. A body starts from rest and covers 120 m in 8th sec. What is the acceleration of the body?
   a. 2 m/s² b. 15 m/s²
   c. 16 m/s2 d. 32 m/s2

5. Two vectors $\vec{a}$ and $\vec{b}$ are parallel to each other when
   a. $\vec{a}.\vec{b} = 0$ b. $|\vec{a} \times \vec{b}| = 0$
   c. $\vec{a} + \vec{b} = 0$ d. $\vec{a} - \vec{b} = 0$

6. The impurity atom with which pure silicon is doped to make N-type semiconductor is
   a. phosphorous b. Indium
   c. Aluminium d. Gallium
7. If ω_1 and ω_2 are dispersive power of lenses of focal lengths f_1 and f_2 respectively. What is the condition for achromatism for two lenses in contact?
   a. $\frac{\omega_1}{\omega_2} = -\frac{f_1}{f_2}$ b. $\frac{\omega_1}{\omega_2} = \frac{f_1}{f_2}$
   c. $\frac{\omega_1}{\omega_2} = -\frac{f_2}{f_1}$ d. $\frac{\omega_1}{\omega_2} = \frac{f_2}{f_1}$
8. Which of the following remains constant during refraction?
   a. velocity b. wavelength
   c. amplitude d. frequency
9. A convex lens produces a real image m times the size of the object. What is the distance of object from the lens?
   a. $(m+1)f$ b. $\frac{m-1}{m}f$
   c. $(m-1)f$ d. $\frac{m+1}{m}f$
10. Which mirror is used in a car?
   a. Convex mirror b. Concave mirror
   c. Plane mirror d. Cylindrical mirror
11. When 10^{14} electrons are removed from a neutral metal sphere, the charge on the sphere is
   a. 16 μC b. -16μC
   c. 32 μC d. -32 μC

Answers

1.	2.	3.	4.	5.	6.	7.	8.	9.	10.	11.
b	a	d	c	b	a	a	d	d	c	a

Group B

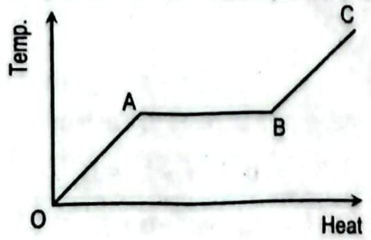
Attempt *ALL* questions. **[8× 5 = 40]**

12. a. Define work energy theorem. [1]
    b. Frictional force is non conservative force. Justify. [1]
    c. A machine gun fires 1000 bullets per minute with a velocity of 200 m/s. If each bullets has mass of 40gm. Calculate the power developed by the machine gun.[3]
    Ans: 13.3 kW

    OR

    a. Define Young's modulus of elasticity. [1]
    b. Steel is more elastic than rubber. Justify. [1]
    c. Calculate the deforming force necessary to cause a wire of cross sectional area 1 mm^2 to increase in length by one-tenth of one percent if the Young's modulus for the wire is 12×10^{10} N/m^2. [3]
    Ans: 120 N

13. a. State and explain Newton's law of cooling. [2]
    b. A solid is heated at constant rate. The variation of temperature with heat input is shown graphically.

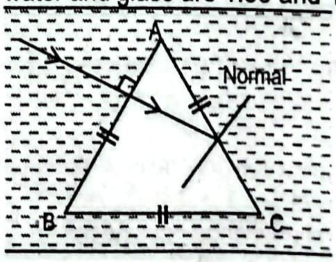


    i. What do OA and AB indicates in the figure? [1]
    ii. What is represented by the slope of BC? What is the reason behind the zero slope during the change of state? [2]

14. a. What is the value of coefficient of Cubical expansivity of a liquid while freezing? How? [2]
    b. Nepali Five-rupee coin has a diameter 2.5 cm at 20°C. The coin is made of a metal alloy (mostly Zinc) for which coefficient of linear expansion is 2.6×10^{-5} K^{-1}. What would be its diameter on a hot day in Bharatpur at 40°C? [3]
    Ans: 2.5013 cm

15. a. Two identical sizes rods made with different materials are kept in same environment.
       i. What can be the ratio of rate of heat flow by conduction? [1]
       ii. Sketch the nature of graph between rate of heat flow and thermal conductivity? [1]
    b. Calculate the temperature at which the root mean square of nitrogen molecules will be equal to 8 kms^{-1}. [Molecular weight of nitrogen=28, R = 8.31 J/kgK] [3]
    Ans: 7.2×10^4 K

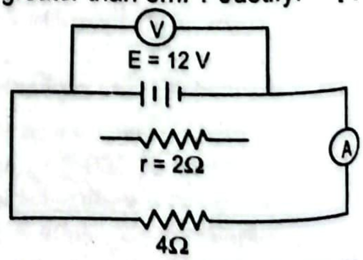
16. a. What is minimum deviation? How can you achieve this condition? [2]
    b. A ray of light is incident on one face of equilateral prism normally. Prism is immersed in water as shown in figure. Predict whether the light ray refract or reflect on the face AC with calculation. (refractive index of water and glass are 1.33 and 1.5 respectively) [3]



    OR

    a. Define lateral shift. Write its expression. [2]
    b. Sketch a graph for the variation of lateral shift with the angle of incident in a rectangular glass slab. [1]
    c. Calculate the critical angle for water – glass surface. (Refractive indices of water and glass are 1.3 and 1.5 respectively). [2]
    Ans: 62.5°

17. a. Define Internal resistance of a cell. [1]
    b. Can p.d of a cell ever greater than emf ? Justify. [2]
    c. For a circuit shown in figure, calculate the voltmeter and ammeter reading.[3]



    Ans: I = 2A, V = 8V.

18. a. Define one electron volt. [1]

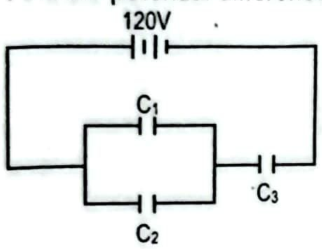
b. The potential difference between two points is 10 volts. What does it mean? [1]

c. A charged oil drop remains stationary when situated between two parallel horizontals metal plates 25 mm apart and a potential difference of 1000 V is applied to the plates. Find the charge on the drop if it has a mass of 5×10^{-15} kg. [3]

Ans: 1.25×10^{-18} C

19. a. State and explain Gauss theorem. [2]

b. In the figure C_1 = 10 μF, C_2 = 20 μF and C_3 = 15 μF. Find the potential difference across capacitor C_3. [3]



Ans: 80 V

Group C

Attempt all questions. [3×8=24]

20. a. Define projectile motion. [1]

b. Show that the path of projectile fired horizontally from the top is parabolic. Also, find the expression for velocity of projectile at any instant. [3]

c. At what point velocity and acceleration of above mentioned projectile are perpendicular? [1]

d. A projectile is fired with a speed of 30 m/s in a direction 30° above horizontal.

Calculate:

i. Position of projectile after time t=3sec

ii. The horizontal range [3]

Ans: (i) 77.94, – 90 (ii) 77.94 m

OR

a. Define impulse. How it is related with linear momentum? [1+1=2]

b. State and prove the principle of conservation of linear momentum. [3]

c. An elevator has a mass of 4000 kg, if the tension in the supporting cable passing through frictionless pulley is 4.8×10^5 N then find the acceleration. Starting from rest, how far does it move in 3 second?[3]

Ans: 495 m

21. a. Which force provides required centripetal force for a satellite to revolve around the earth? [1]

b. Derive an expression for variation of acceleration due to gravity with depth from the surface of earth. [3]

c. Sketch a graph showing the variation of acceleration due to gravity inside and outside of earth. [1]

d. An artificial satellite revolves round the earth in 2.5 hours in a circular orbit. Find the height of the satellite above the earth assuming earth as a sphere of radius 6370 km. [3]

Ans: 2982 km

22. a. Define isotopes. Give any two examples of it. [2]

b. All the nuclei have nearly the same density. Justify. [2]

c. A fusion reaction is more energetic than fission reaction. Justify. [1]

d. Calculate the binding energy per nucleon of $_3Li^7$ from the given data below.

(mass of $_3Li^7$ = 7.01435 amu, mass of $_0n^1$ = 1.00867 amu and mass of $_1H^1$ = 1.00728 amu) [3]

Ans: 5.6 MeV/Nucleon

Set 6: HISSAN 2079

Group A

Multiple Choice Question Attempts all questions.
Rewrite the correct option in your answer sheet.[11×1=11]

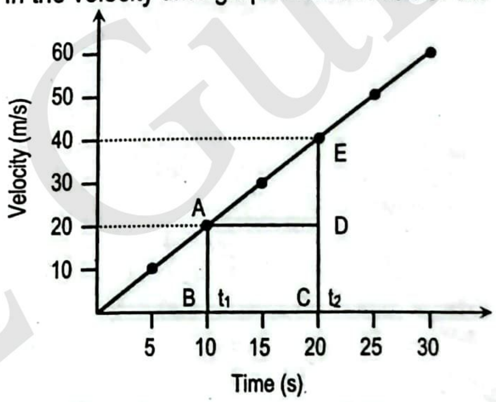
1. The dimensional formula of stress is same as that of

a. Force | b. Impulse
c. Pressure | d. energy

2. Two displacement vectors 3m and 4m and their dot product is 6m. The angle between them is……

a. 30° | b. 60°
c. 50° | d. 40°

3. In the velocity time graph acceleration of the body is



a. 1.70 m/s² | b. 2.00 m/s²
c. 1.90 m/s² | d. 1.65 m/s²

4. A wire of length 3m and area of cross section 1×10^{-6} m² has a mass of 15 kg hanging on it. What is the extension produced? [$Y = 2\times10^{11}$ N/m²]

a. 3 mm | b. 2.25 mm
c. 3.25 m | d. 2.5 mm

5. A wooden block is on the frictionless table inclined at an angle of 30° to the horizontal. The acceleration of block down the plane is

a. 9.8m/s² | b. 10m/s²
c. 5m/s² | d. 2.5m/s²

6. Which mirror is preferred for shaving and make up

a. Plane mirror | b. Convex mirror
c. Parabolic mirror | d. Concave mirror

7. Calculate the velocity of the light ray in the medium, if the critical angle of medium is 30°.

a. 2×10^8 m/s | b. 1.5×10^8 m/s
c. 0.75×10^8 m/s | d. 3×10^8 m/s

8. Rainbow is an example of which phenomenon?

a. Refraction and scattering
b. Refraction and total internal reflection

c. Dispersion and reflection
d. Dispersion and total internal reflection

9. Two thin lenses of focal length 15 cm and 45 cm are placed in contact. The power of the combination is
a. 9D b. 5D
c. 7D d. 10D

10. Energy stored in a charged capacitor of capacitance C and having p.d of 10V is
a. 5C b. 50C
c. 500C d. 0.5C

11. Unit of electric flux is.....
a. Coulomb b. Coulomb m^2
c. Volt d. N m^2 / Coulomb

Answers										
1.	2.	3.	4.	5.	6.	7.	8.	9.	10.	11.
c	b	b	b	c	d	b	d	d	d	b

Group B

Attempt ALL questions. **[8×5=40]**

12. A cricket player hits a cricket ball with certain angle θ with horizontal with velocity v.

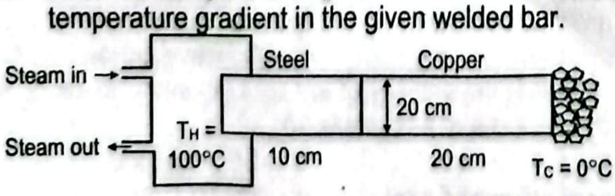
a. What kind of motion is this? [1]
b. Find the time required by cricket ball to reach the ground if could not be catched by another player. [2]
c. You toss a ball from the window 8m above the ground. When the ball leaves your hand, it is moving at 10 m/s at an angle of 20^0 below the horizontal. How far horizontally from the window will the ball hit the ground? [2]

Ans: 9.12 m

OR

a. An object is moving in circular motion, explain why the object is accelerating even though it is moving at constant speed. [1]
b. Derive the expression for centripetal force of an object moving in circular motion. [3]
c. Give two applications of centripetal force in daily life.[1]

13. a. What is thermal conductivity of substance. Give its SI unit. [2]
b. What is temperature gradient? Find its value of temperature gradient in the given welded bar. [2]



Ans: 800.2° c/m, 1034° c/m

c. Wooden home is warmer than cemented home. Why? [1]

14. a. What is absolute zero temperature? [1]
b. Find the expression of pressure exerted by gas in a cubical vessel. [3]
c. What is Avogadro number? [1]

15. a. What is Prism? Draw its 3-d diagram. [2]
b. A glass prism of angle of prism A and refraction index 1.5 produces the angle of minimum deviation equal to $40°$. Calculate the value of angle of prism. [3]

Ans: 62.8°

OR

a. Write down lens makers formula. Name the physical quantities evolved in the formula. [2]
b. What happens to the focal length of lens if it is dipped in water? [1]
c. Write down the applications of lens in human life. [2]

16. a. Does specific heat capacity depend upon mass? [1]
b. The difference between the length of a certain brass rod and that of a steel rod is claimed to be constant at all temperatures. Is it possible? [1]
c. A brass rod of length 50cm and diameter 3mm is joined to steel rod of the same length and diameter. What is the change in length of the combined rod at 250°C, if the both rods are at 40°C initially? Coefficient of linear expansion of brass and steel are 2.1×10^{-5} °C^{-1} and 1.2×10^{-5}°C^{-1} respectively. [3]

Ans: 0.346 cm

17. a. State Coulomb's law with its mathematical expression. [2]
b. Find the expression of electric field intensity of plane sheet of charge using Gauss theorem. [3]

18. a. What is electric potential? [1]
b. Derive the formula for the potential at a point due to a point charge. [3]
c. What are the factors affecting capacitance of a capacitor? [1]

19. a. State Ohms law. Why it is not universal law? [2]
b. A 3-volt battery of internal resistance 1Ω is connected to the two resistors of 3Ω and 6Ω in parallel. Find the ratio of the heat developed in the resistors. [3]

Ans: 2:1

Group C

Attempt all questions. **[3 × 8=24]**

20. a. What is escape velocity? [1]
b. Find the expression of escape velocity of earth. [3]
c. A student said that escape velocity of earth and moon are equal. Do you agree with him? [1]
d. Find the height of Geostationary satellite of earth. [M = 6 x 10^{24} kg, R = 64000km] [3]

Ans: 36000 km

OR

a. What is elastic collision? Is it possible in our daily life?[2]
b. Prove that in one dimensional elastic collision relative velocity of approach is equal to relative velocity of separation. [3]

c. A body of mass 2 kg, initially at rest, is acted upon simultaneously by two forces, one of 4N and the other of 3N acting at right angles to each other. Find the kinetic energy of the body after 20s. [3]

21\. a. Differentiate between static and dynamic friction. [2]

b. State and prove principle of conservation of linear momentum using Newton's laws of motion. [3]

c. A motor car of mass running at the rate of 7 m/s can be stopped by its brakes in 10 m. Prove that the total resistance to the car's motion, when the brakes are on, is one quarter of the weight of the car. [3]

Ans:2500 J

22\. a. Why density of all nuclei is same? Justify. [2]

b. What are binding energy and packing fraction? [2]

c. Find mass defect and binding energy of ${}_{26}Fe^{56}$. Mass of ${}_{26}Fe^{56}$ = 55.934939 a.m.u., m_p = 1.00727 a.m.u. and m_n = 1.00866 a.m.u. [1amu = 931MeV] [3]

Ans: 0.514 amu, 478.7 MeV

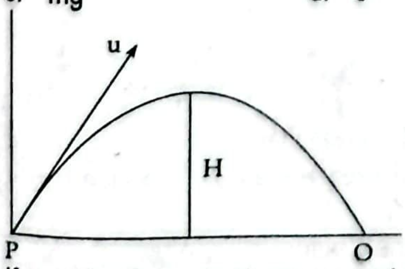
Set 7

Attempt All questions.

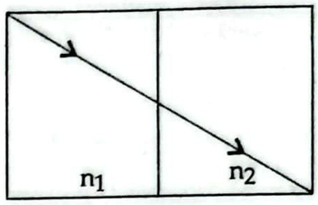
Group 'A'

Circle the best alternative to the following questions [11×1=1]

1. What is the vertical component of the force 5N acting on a particle along a direction making an angle of 60°?
   a. 3 N b. 2.5 N
   c. 10 N d. 4 N
2. NC^{-1} has the same dimension as
   a. Volt meter b. Farad meter
   c. Farad/meter d. Volt/meter
3. If KE of a body decreases by 10% then decrease in linear momentum of that body
   a. 2% b. 10%
   c. 5% d. 15%
4. The direction of two force 6 N and 2N action on a body of mass 2 kg is same, the minimum acceleration of the body cannot be less than;
   a. 2 m/s^2 b. 2.5 m/s^2
   c. 3 m/s^2 d. 4 m/s^2
5. When a body projected from P to O with velocity u m/sec, the work done in these points from P to O is (air resistance is neglected)
   a. ½ mu^2 b. 2 mg
   c. mg d. 0



6. If combined power of convex and concave lens is 40 D. The power of convex lens is
   a. 1 cm^{-1} b. 100 cm^{-1}
   c. 10 cm^{-1} d. 100 cm
7. Total internal reflection occour when the light passes
   a. Denser to rearer b. Rearer to denser
   c. The μ is same d. None
8. Optical fibre is based on phenomenon.
   a. Total internal reflection
   b. Refraction
   c. Scattrering
   d. Dispersion
9. A light ray is passed from one medium of RI(n_1) to another medium RI(n_2) as shown in fig. The correct relation between n_1 and n_2 is



   a. $n_1 > n_2$ b. $n_1 < n_2$
   c. $n_1 = n_2$
   d. There is no relation between n_1 and n_2
10. The space around stationary charge has
   a. Electric field only b. Magnetic field only
   c. Localised electric as well as magnetic field
   d. Electric and magnetic fields that are radiated
11. The percentage of mass which changes into energy during fission is in the order of:
   a. 10% b. 1%
   c. 0.4% d. 0.1%

Answers

1.	2.	3.	4.	5.	6.	7.	8.	9.	10.	11.
b	d	c	a	d	c	a	a	c	a	d

Group 'B'

Answer the following questions. [8×5=40]

12\. a. How many components can be realized of a vector? [1]

b. Show there are two angles of projection of a projectile for the same horizontal range. [2]

c. Why is it necessary to bend knees while jumping from greater height? [2]

OR

a. State law of conservation of energy. [1]

b. Show that final kinetic energy of a body is equal to the sum of initial kinetic energy and work done on the body. [3]

c. Can a body have energy without momentum? [1]

13\. a. Why Zeroth law is named so? [1]

b. Why is mercury used commonly as a thermometric substance? [2]

c. Does the coefficient of linear expansion depend on length? Explain. [2]

14\. a. From what height should a block of ice be dropped in order that it may melt completely? (Specific latent heat of ice (L_i) = 336000 J kg^{-1}) [3]

Ans: 3.43×10^4 m

b. Why fusion of ice makes a line (or locus) in p–v diagram? [2]

15\. a. Define steady state. [1]

b. The silica cylinder of a radiant wall heater is 0.6 m long and has a radius 6 mm. If it is rated at 1.5 kw estimate its temperature when operating. [The Stefan constant, $\sigma=6\times10^{-8}$ $wm^{-2}k^{-4}$] [3]

Ans: 1025 k

c. Out of three mechanisms of transmission of heat, which one is fastest? [1]

16. a. What size of mirror is required to see full image? [1]

b. How long will the light take in travelling a distance of 500 m in water refractive index of water is 1.33 and velocity of light in vacuum is 3×10^8 m/s. [2]

Ans: 2.2×10^{-6} s

c. A spherical mirror is cut in half horizontally will an image be formed by the bottom half of the mirror? How? [2]

OR

a. "A concave mirror is called converging mirror." Why?[1]

b. State one daily application of convex mirror. [1]

c. A ray of light is incident at 60° in air on an air-glass plane surface. Find the angle of refraction in the glass (μ for glass = 1.5). [3]

Ans: 35.3°

17. a. What is the force between two charges of equal magnitude of 1 C when placed 1 m distance? [1]

b. Near the earth's surface, the electric field in the open air has magnitude 150 N/c and is directed down towards the ground. If this is regarded as being due to a large sheet of charge lying on the earth's surface, calculate the charge per unit area in the, what is the sign of the charge? [3]

Ans: 2.66×10^{-1} c/m², negative

c. Define electric flux. [1]

18. a. Can two equipotential surfaces intersect? Justify your answer. [2]

b. The surface charge density of an infinite charged sheet is 1μ c/m². How far apart are the equipotential surfaces whose potentials differ by 10 volt? [3]

Ans: 1.78×10^{-4} m

19. a. What is one unit electricity? [1]

b. A 50 A.h. battery can supply a current of 50A for 1.0 h. What total energy can be supplied by a 12v. 60 A.h battery if its internal resistance is negligible? [2]

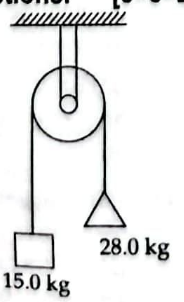
Ans: 2.6×10^{6} J

c. Can terminal potential difference by greater than emf of a cell? [2]

Group C

Given long answer in the following questions. [3×8=24]

20. At wood's machine: A 15.0 kg load of brick hang's from one end of a rope that passes over a small, frictionless pulley. A 28.0 kg counter weight is suspended from the other end of the rope, as shown in figure. The system is released from the rest.

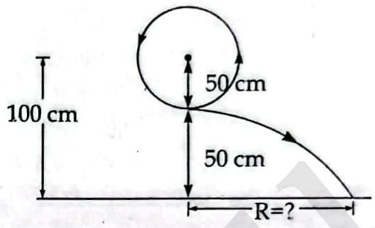


a. Draw two free body diagrams, one for the load of bricks and one for the counter weight. Why free body diagram is very important in the calculation of force.[2+1]

b. What is the magnitude of the upward acceleration of the load of bricks? What is the tension in the rope while the load is moving? [3]

c. How does the tension compare to the weight of the load of bricks to the weight of the counter weight? [2]

21. A stone of mass 500 g is attached to a string of length 50 cm which will break if the tension in it exceeds 20 N. The stone is whirled in a vertical circule, the axis of rotation being at a height of 100 cm above the ground as shown in figure.



a. In what position is this break most likely to occur and at what angular speed? [3]

Ans: 7.75 rad/s

b. Where will the stone hit the ground? [2]

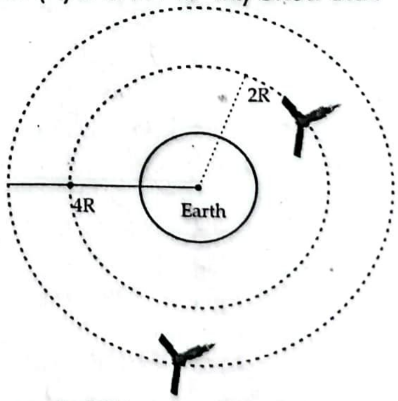
Ans: 1.24 m

c. What would be the centripetal acceleration and centripetal force? [3]

Ans: 30.03 ms⁻¹, 15.01 N

OR

A satellite X moves round the earth in a circular orbit of radius R. Another satellite Y of the same mass moves round the earth in a circular orbit 4R. Shown in the figure. (Use the following information if the mass of satellites are 1000 kg, velocity of first satellite is 6.2 kms⁻¹, second satellite is 3.1 kms⁻¹ radius of X satellite (R) is 6.4×10^6 m.) Show that:



a. The speed of X is twice that of Y. [3]

b. The K.E. of X is greater than of Y and the P.E. of X, is less than of Y. [3]

c. As X or Y has the greater total energy. [2]

22. a. What is nuclear fusion? How energy is released in nuclear fusion reaction? [3]

b. Assuming that about 200 MeV energy is released per fission of $_{92}U^{235}$ nuclei, what would be the mass of

U^{235}, consumed per day in the fission reactionfo power 1 MW approximately? [3]

Ans: 1.05 g

c. Why is the mass of a nucleus slightly less than the mass of consistituent particles? [2]

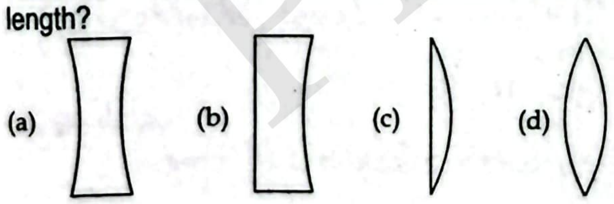
Set 8

Attempt All questions.

Group 'A'

Circle the best alternative to the following questions [11×1=1]

1. Angular momentum is
   a. A scalar vector b. An axial vector
   c. A polar vector d. A displacement vector
2. The dimensional formula of impulse is
   a. $[MLT^{-2}]$ b. $[MLT^{-1}]$
   c. $[ML^2T^{-1}]$ d. $[M^2LT^{-1}]$
3. Kinetic energy of a body of mass 10g and momentum 500 g cm/s is equal to
   a. 1.25×10^3 ergs b. 1.25×10^4 ergs
   a. 1.25×10^3 J b. 50,000 ergs
4. What of the following is always conserved in collusion?
   a. Momentum b. Power
   c. Kinetic energy d. Torque
5. When an aeroplane moves with 600 km/h due east & return with 400 km/hr then what is average speed for entire journey?
   a. 240 b. 480
   c. 500 d. 720
6. If sky is seen from moon's surface, it will appear:
   a. blue b. black
   c. white d. red
7. A lens behave as a converging lens in air and diverging lens in water. The refractive index of lens –material is
   a. Equal to air
   b. Equal to water
   c. More than air & less then water
   d. More than water
8. If the foar lenses shown below are made of the same material, which lens has the shortest positive focal length?



9. What is the magnification when the object placed at 2f from the pole of a convex mirror?
   a. –1/2 b. –2/3
   c. –1 d. –3/2
10. A hollow sphere of charge does not produce an electric field at any
   a. inner point b. Outer point
   c. Surface point d. None of the these
11. The density of mass number A proportional to
   a. A^3 b. $A^{1/3}$
   c. A^1 d. A^0

Answers

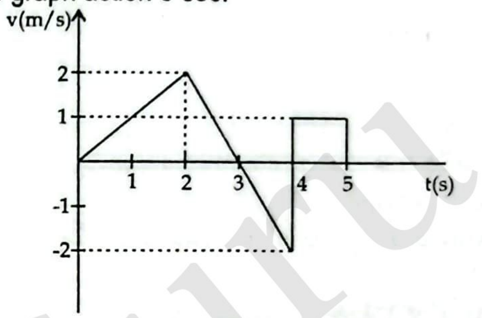
1.	2.	3.	4.	5.	6.	7.	8.	9.	10.	11.
b	b	b	a	b	b	c	d	a	a	d

Group 'B'

Answer the following questions. [8×5=40]

12. a. A swimmer wants to reach a point just opposite of a bank of a river, how should he dive to achieve his goal? [2]

b. The velocity of time graph is shown in the figure below. Calculate the displacement and distance from the graph action 5 sec. [3]



Ans: Displacement = 3 m, distance = 3 m

OR

a. Which force is responsible to revolve a body in a circular path? [1]

b. A spherical ball contracts in volume by 0.1%, when subjected to a normal uniform pressure of 100 atmosphere. Calculate volume strain. [2]

Ans: 10^{-3}

c. Comment on the statement "Sharper the curve, more is the bending." [2]

13. a. What are the units and dimensions of linear expansivity and cubical expansivity? [1]

b. A glass vessel of volume 50 cm^3 is filled with mercury and is heated from 20°C to 60°C. What volume of mercury will overflow? [Given: Linear expansivity of glass (α_g) = 1.8×10^{-6} K^{-1} and volume expansivity of mercury (γ_m) = 1.8×10^{-4} K^{-1}] [3]

Ans: 0.35 cm^3

c. Why gas thermometers are more sensitive than mercury thermometers? [1]

14. a. State and explain Stefan's law of black body radiation. [3]

b. Calculate the rate of loss of heat through a glass window of area 200 cm^2 and thickness 0.5 cm when temperature inside is 35°C and outside –5°C, coefficient of thermal conductivity of glass is 2.2×10^{-3} cal $s^{-1}cm^{-1}K^{-1}$. [2]

Ans: 35.2 cal s^{-1}

15. a. What do you mean by Brownian motion? [1]

b. The total lungs volume for a typical physical student is 6×10^{-3} m^3. A physics student fills her lungs with air at an absolute pressure of 1.01×10^5 Nm^{-2}. Then, holding her breath, she compresses her chest cavity, decreasing her lungs volume to 5.7×10^{-3} m³. What is the pressure of the air in her lungs then? Assume that the temperature of the air remains constant? [3]

Ans: 1.05 atm

c. State Charles' law. [1]

16. a. Why does air bubble shine in the water? [2]
b. At what condition, a concave mirror forms virtual and magnified image? [1]
c. What are grazing incidence and grazing emergence?[2]

OR

a. Define principal focus. Why is it called so? [2]
b. (i) Define power of lens. [1]
(ii) The radii of curvature of double concave lens are 30 cm and 60 cm. The refractive index of its glass is 1.5. Calculate the focal length of the lens. [2]

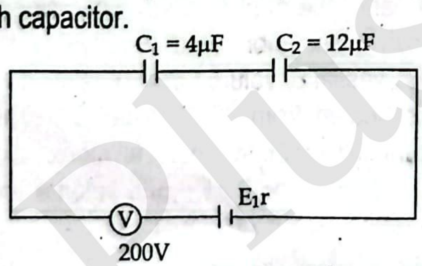
Ans: – 40 cm

17. a. What are the properties of electric lines of force? [2]
b. Derive electric field intensity expression due to a point charge. [3]

18. a. What are equipotential surfaces? Why are they called so? [2]
b. An isolated conducting spherical shell of radius 0.10 m, in vacuum, carries a positive charge of 1.0×10^{-7} C. Calculate (i) the electric field intensity, (ii) the potential, at a point on the surface of the conductor.[3]

Ans: 9×10^4 Vm^{-1} (or NC^{-1})

19. a. Define 1 farad. [1]
b. Two capacitances are connected in a series and an identical pair is connected in parallel. They are connected to the same voltage source. Which pair would be more dangerous to handle? [2]
c. Calculate the equivalent capacitace and charge on each capacitor. [2]



Ans: 3×10^{-6}F, 16×10^{-6} F, 6×10^{-4} C

Group C

Given long answer in the following questions. [3×8=24]

20. a. A railway truck of mass 4×10^4 kg moving at a velocity of 3 ms^{-1} collides with another truck of mass 2×10^4 kg which is at rest. The coupling joins and the trucks move off together. What fraction of the first truck's initial kinetic energy remains as kinetic energy of the two trucks after the collision? [3]

Ans: 2:3

b. A loaded grocery cart is rolling across a parking lot in a strong wind. You apply a constant force $\vec{F} = (30\ \text{N})\ \hat{i} - (40\ \text{N})\ \hat{j}$ to the cart as it undergoes a displacement $\vec{s} = (-9.0\ \text{m})\ \hat{i} - (3.0\ \text{m})\hat{j}$. How much work does the force you apply do on the grocery cart? [2]

Ans: 150 J

c. A uniform ladder rests on a rough horizontal ground, leaning against a smooth vertical wall. Weight of the ladder is 300 N and a man weighing 500 N stands on one quarter of its length from the bottom. Ladder makes 30° to the horizontal. Find the reaction at the wall and the total force on the ground. [3]

Ans: 931.06 N

21. a. When a particle is revolved round in a horizontal circle. What physical quantities remain constant? [2]
b. A clock has a second hand 1.5 cm long. Calculate (i) speed of the tip of second hand. The velocity of the top at t = 0 to t = 30 s (ii) change in velocity from t = 0 to t = 30 s. [3]

Ans: 0.16 cms^{-1}, 0.16 cms^{-1} and 0.32 cms^{-1}

c. A mass of 1 kg is attached to the lower end of a string 1 m long whose upper end is fixed. The mass is made to rotate in a horizontal circle of radius 60 cm. If the circular speed of the mass is constant, find the tension in the string and the period of motion. [3]

Ans: 12.5 N, 1.78 s

OR

a. How does modulus of elasticity change with the rise of temperature? [2]
b. A wire 2 m long and cross-sectional area 10^{-6} m² is stretched 1 mm by a force of 50 N in the elastic region. Calculate (i) the strain (ii) the Young modulus (iii) the energy stored in the wire. [3]

Ans: (i) $\frac{1}{200}$ (ii) 50×10^6 (iii) 0.025 J

c. An artificial satellite revolves round the earth in 2.5 hours in a circular orbit. Find the height of the satellite above the earth assuming earth as a sphere of radius 6370 km. [3]

Ans: 3041 km

22. a. State Hubble's law and give its significance. [2]
b. Find the distance of the galaxy moving with speed 1.55×10^7m/s from the earth, according to the Hubble law.
($H_0 = 17 \times 10^{-3}$ ms^{-1}/ly). [3]

Ans: 87.032×10^{23} m

c. Explain the constituents of the universe. [3]

Set 9

Attempt All questions.

Group 'A'

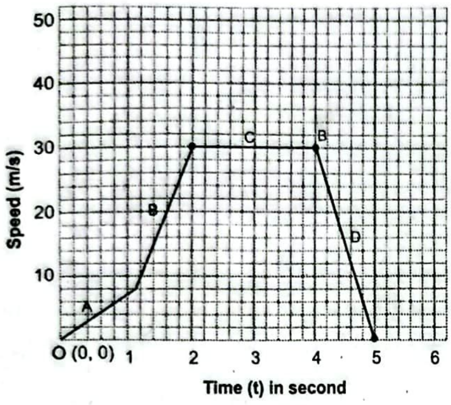
Circle the best alternative to the following questions [11×1=1]

1. A boy walks uniformly along the sides of a rectangular park with dimension 400 m × 300 m starting from one corner to the other corner diagonally opposite. Which of the following statement is false?

a. His displacement is 700 m
b. His displacement is 500 m
c. He has travelled a distance of 700 m
d. His velocity is not uniform throughout the walk

2. The dimensional formula of coefficient of viscosity is
a. $[ML^{-1}T^{-1}]$ b. $[ML^0T^0]$
c. $[ML^{-2}T^{-1}]$ d. $[ML^{-1}T^{-2}]$

3. The speed-time graph represents a motorcycle journey. In which part of the graph is the acceleration equal to zero?

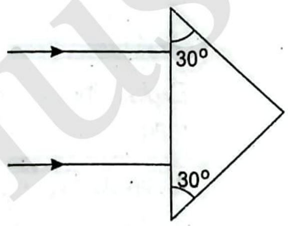


a. A b. B
c. C d. D

4. The product of moment of inertia and angular acceleration gives
a. Linear momentum b. Angular momentum
c. Torque d. Force

5. In an experiment to determine the Young modulus of elasticity of the material of wire and the suspended load are both doubled. The ratio of stress to strain.
a. Becomes double
b. Remains unchanged
c. Becomes four times by spare
d. Becomes sixteen times

6. Two parallel light rays are incident at one surface of a prism as shown in figure. The prism made of glass of refractive index 1.5. The angle between the rays as they emerge is nearly equal to



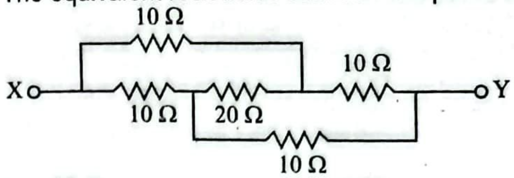
a. 19° b. 45°
c. 49° d. 37°

7. Object is placed at a distance of 40 cm from screen and a convex lens is placed exactly between then for the image to be obtained on screen, focal length of lens should be
a. Less than 40 cm b. Between 20 & 40 cm
c. More than 40 cm d. 10 cm

8. Power of lens is measured in
a. Diopter b. Calorie
c. Gram d. Centimeter

9. Which of the following remains constant during refraction?
a. Velocity b. Wavelength
c. Amplitude d. Frequency

10. The equivalent resistance between two points X and Y is



a. 10 Ω b. 20 Ω
c. 22 Ω d. 60 Ω

11. Holes are majority charge carriers in
a. N-type semiconductors. b. ionic solids.
c. P-type semiconductors. d. metals.

Answers										
1.	2.	3.	4.	5.	6.	7.	8.	9.	10.	11.
a	a	b	c	b	d	d	a	d	a	c

Group 'B'

Answer the following questions. **[8×5=40]**

12. a. State parallelogram law of vector addition. [1]

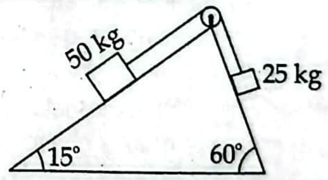
b. A football is kicked vertically upward from the ground and a student gazing out of the window sees it moving upward past her at 5.00 ms^{-1}. The window is 12.0 m above the ground. You may ignore air resistance. How high does the football go above the ground? [2]

Ans: 13.2 m

c. A bomb is to be dropped from a moving helicopter on the ground. Explain how can it hit the target? [2]

OR

a. Two blocks connected by a cord passing over a small frictionless pulley rest on frictionless planes as shown in the diagram. (i) Which way will the system move? (ii) What is the acceleration of the system? (iii) What is the tension in the cord? [3]



Ans: (i) 25 kg move down, (ii) 1.16 ms^{-2} (iii) 187. 46 N

b. Why a wrench of longer arm is preferred in comparison to a wrench of shorter arm? [2]

13. a. What is the difference between a real and ideal gas? [1]

b. You are designing an electronic circuit element made of 23 mg of silicon. The electronic current through it adds energy at the rate of 7.4 mW = 7.4×10^{-3} J/s. If your design doesn't allow any heat transfer out of the element, at what rate does its temperature increase? The specific heat of silicon is 705 J/kg K. [3]

Ans: 0.46 k/s

c. Differentiate between fusion curve between H_2O and CO_2. [1]

14. a. Define the relation of kinetic energy of a gas molecule with the absolute temperature. [3]

b. Which has more molecules: A kilogram of hydrogen on a kilogram of oxygen? [2]

15. a. Why snow is a better heat-insulator than ice? [1]
b. Calculate the rate of loss of heat through a glass window of area 200 cm^2 and thickness 0.5 cm when temperature inside is 35°C and outside –5°C, coefficient of thermal conductivity of glass is 2.2×10^{-3} cal $s^{-1}cm^{-1}K^{-1}$. [3]
Ans: 35.2 cal s^{-1}
c. Heat is generated continuously in an electrical heater but its temperature becomes constant after sometime. Why? [1]

16. a. Define concave mirror and state one daily application of it. [2]
b. A certain projecter uses a concave mirror for projecting an objects image that is a 5 times bigger than the object and the screen is 5 times bigger than the object and screen is 5 m away from the mirror as in figure (i)
i. Give the reason why is the image larger than the object? [1]

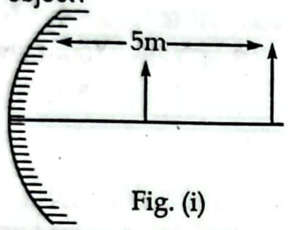


Fig. (i)

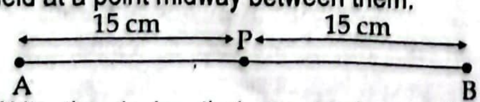
ii. Calculate the focal length of the mirror. [2]
Ans: 2.5 m

OR

a. Does rainbow a pure spectrum? [1]
b. The sun appears red at sunset and sunrise, why? [2]
c. The deviation produced by a flint glass prism for violet and red light rays are 3.25° and 3.10° respectively. Find the angular dispersion. [2]
Ans: 0.15°

17. a. Vehicles carrying flammable material and running on rubber tyres always drag a chain along the ground, why? [2]
b. Describe the effect of permittivity on force between two charges. [3]

18. a. "Nothing happens to a bird standing on high power line but a man gets fatal shock by touching to it," why? [2]
b. Two point charges of magnitude 1.0×10^{-8} C and 2.0×10^{-8} C are 30 cm apart in air. Find the electric field at a point midway between them. [3]

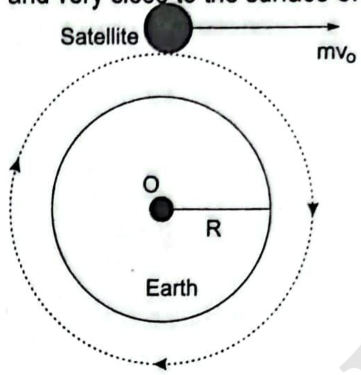


19. a. Write the device that can produce uniform electric field. [1]
b "No two equipotential surfaces intersect." why? [2]
c. What is the potential gradient between two parallel plane conductors when their separation is 20 mm and a p.d. of 400V is applied to them? [2]
Ans: 2×10^4 vm^{-1}

Group C

Given long answer in the following questions. [3×8=24]

20. a. The escape velocity of earht's is 11.2 Km sec^{-1}. What does it means? [2]
b. From figure, consider the earth to be a perfect sphere of mass 'M' and radius 'R' with centre O. Find the expression of orbital velocity of satellite at height 'h' and very close to the surface of earth. [3]



c. Show that escape velocity of as body from the earth's surface is $\sqrt{2}$ times its velocity in a circular orbit just above the earth's surface. [2]
d. State the necessary conditions for a satellite to be stationary and hence find its height. [2]

21. a. According to Newton's law of gravitation, the apple and the earth experience equal and opposite forces due to gravitation. But it is the apple that falls towards the earth and not the earth to the apple. Why? [2]
b. Two people carry a heavy electric motor by placing it on a light board 2.00 m long. One person lifts one end with a force of 400 N, and the other lifts the opposite end with a force of 600 N. What is the weight of the motor and where along the board is its centre of gravity located? [3]
c. Describe the working principle of GPS. [3]
Ans: 1000 N, 1.2 m

OR

a. Explain the significance of the banking of a curved road. [2]
b. A playing record of radius 15.0 cm revolves with $33\frac{1}{3}$ rev/min. Two coins are placed at 5 cm and 10 cm away from the centre of record. If the coefficient of friction between the coin and record is 0.15, so that coins revolve with the record. Find the position of coins. [3]
Ans: $r \leq 12$ cm
c. Derive an expression for the banking angle. [3]

22. a. "Heavy nuclei split into lighter nuclei, by a process called fission." Why? [2]
b. Diameter of Al^{27} nucleus is D_{Al}. How can one express the diameter of Cu^{64} in terms of D_{Al}? Explain. [3]
c. A neutron is absorbed by a $_3Li^6$ nucleus with subsequent emission of an α-particle. Write the corresponding nuclear reaction and calculate the energy released in the reaction. (Given; Mass of

neutron = 1.008665 amu, Mass of ${}_3Li^6$ = 6.015126 amu, Mass of ${}_2He^4$ = 4.002603 amu, 1 amu = 931 MeV) [3]

Ans: 4.784 Mev

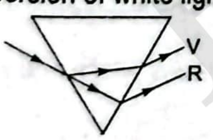
Set 10

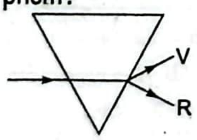
Attempt All questions.

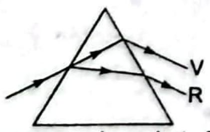
Group 'A'

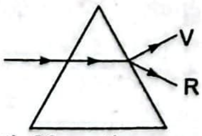
Circle the best alternative to the following questions [11×1=1]

1. Any vector in an arbitrary direction can be replaced by two or three vectors
   a. Perpendicular to each other and have the original vector as the resultant
   b. Parallel to each other and have the original vector as their resultant
   c. Arbitrary vectors which have original vectors as their resultant
   d. It is impossible to resolve a vector
2. Dimension of universal constant of gravitation .
   a. $[ML^{-1}T^{-3}]$ b. $[M^{-1}L^3T^{-2}]$
   c. $[ML^{-3}T^{-1}]$ d. $M^2L^2T^{-2}]$
3. If an object of mass 'm' moving in circular path with uniform speed 'v' which of the following change occurs in half circle.
   a. Kinetic energy changes by $\frac{mv^2}{2}$
   b. Kinetic energy changes of $\frac{m^2v^2}{4}$
   c. Momentum changes by 2 mv
   d. Momentum doesn't changes
4. A body of mass 200 gm is thrown upwards at 30 m/s. What is the total energy of body?
   a. 50 J b. 40 J
   c. 100 J d. 90 J
5. The force of friction is 50 N in a body weighting 50 kg. Find the coefficient of friction.
   a. 0.5 b. 2.5
   c. 0.102 d. 0.20 L
6. Which of the following diagrams shows correctly the dispersion of white light by a prism?

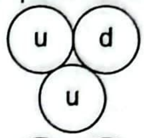
a. 

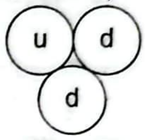
b. 

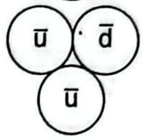
c. 

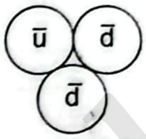
d. 

7. The convex lens has focal length 20 cm the power is
   a. – 5.0 D b. + 5.0 D
   c. + 0.05 D d. – 0.05 D
8. To remove the achromatic aberration, the combination of lenses should be such that
   a. $F_R + F_V = 0$ b. $F_R < F_V$
   c. $F_R < F_V$ d. $F_R - F_V = 0$
9. Angle of minimum deviation for a prism of refractive index 1.5 is equal to the angle of prism. Then angle of prism is
   a. 62° b. 41°
   c. 82° d. 31°
10. The capacitance of a parallel plate capacitor does not depend upon
    a. Area of plates
    b. Medium between the plates
    c. Distance between the plates
    d. Metal of plates
11. The quark combination of antineutron is,

a. 

b. 

c. 

d. 

Answers										
1.	2.	3.	4.	5.	6.	7.	8.	9.	10.	11.
c	b	c	d	c	a	b	d	c	d	d

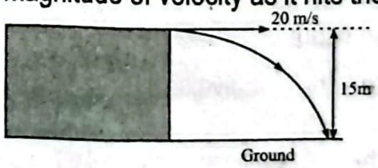
Group 'B'

Answer the following questions. [8×5=40]

12. a. Can an object with constant acceleration reverse its direction? Explain. [2]
    b. Find the angle of projection at which the horizontal range and maximum height of a projectile are equal.[2]
    c. Under what condition will the distance and the displacement of a moving object have the same magnitude. [1]

OR

a. Distinguish between vector and scalar. [1]
b. List out the vectors only among the quantities: Acceleration, force, kinetic energy, mass, power, weight.
c. A stone is thrown horizontally at speed of 20 ms^{-1} from a top of a cliff 15 m high. Air resistance is negligible. Calculate the time of fall and the magnitude of velocity as it hits the ground. [1+2]



Ans: $10\sqrt{7}$ m/sec.

13. a. Why should the bore of a thermometer tube be uniform? [1]
    b. A faulty thermometer has its fixed points marked 2° and 98°. Temperature of a body as measured by the faulty thermometer is 64°. Find the correct temperature of the body on Celsius scale. [3]

Ans: 64.6°C

    c. What is the temperature of vaccum? [1]
14. a. Explain the anomalus expension of water. What is the environmental significance of anomalus expansion of water? [3]

b. Show that $\beta = 2\alpha$ where α and β are linear and superficial expansivities. [2]

15. a. Copper has specific heat capacity 390 Jkg^{-1} K^{-1}. What is its water equivalent? [1]

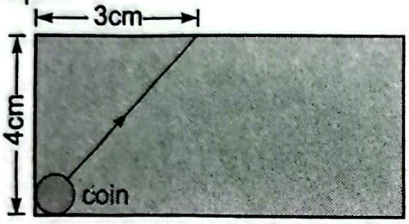
Ans: 10.77 kg

b. Derive the expression for kinetic energy of a gas molecules. [3]

c. Write the formula for combined ideal gas equation. [1]

16. a. What is meant by total internal reflection and write condition for it. [2]

b. A small coin is resting on the bottom of beaker filled with liquid. A ray of light from coin travels up to the surface of the liquid and moves along its surface as shown in fig. Estimate the velocity of light in the liquid. [3]



Ans: 1.8×10^8m/ sec.

OR

a. Why is the power of a lens measured as the reciprocal of its focal length? [1]

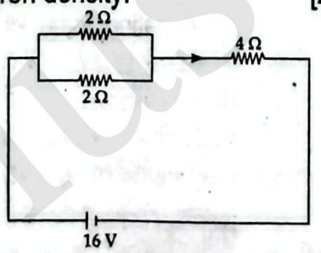
b. On what factors does the focal length of a lens depend? [2]

c. The radii of curvature of biconvex lens are 15 cm and 30 cm and the refractive index of its glass is 1.5. Calculate the focal length of the lens. [2]

Ans: 100 cm

17. a. Define the term 'drift velocity' of charge carriers in a conductor. Obtain the expression for the current density in terms of electron density. [2]

b. A network of resistors is connected to a 16V battery of internal resistance of 1Ω as shown I n the figure.



i. Compute the equivalent resistance of the network. [1]

ii. Obtain the voltage drops across 4 Ω resistor. [2]

Ans: (I) 5Ω (ii) 10.67 V

18. a. How will you arrange 3 capacitors each having the capacity of 2μF, to get a capacitor of capacity 3μF?[2]

b. Three capacitors of 1μF, 2μF and 3μF are connected first in series. The same capacitors are again connect in parallel. Compare their equivalent capacitance. Which one of these combination gives larger value of capacitance? [3]

Ans: 11C

19. a. In a conductor, large number of electrons are free to move in it, but why no current is detected? [2]

b. A "540-W" electric heater is designed to operate from 120 V lines. (a) What is its resistance? (b) What current does it draw? (c) If the line voltage drops to 110 V, what power does the heater take? (Assume that the resistance is constant. Actually, it will change because of the change in temperature.) [3]

Ans: (a) 26.7 Ω (b) 4.5 A (c) 454 W

Group C

Given long answer in the following questions. [3×8=24]

20. a. What kind of energy is stored in spring of a watch? [2]

b. Show that the acceleration of a body moving in a circular path of radius r with uniform speed v is v^2/r. [3]

c. An earth satellite moves in a circular orbit with a speed of 6.2 kms^{-1}. Find the time of one revolution and its centripetal acceleration. [3]

Ans: 2.99 hrs., 3.6 m/sec.

21. a. Why are rubbers used as vibration absorber? [2]

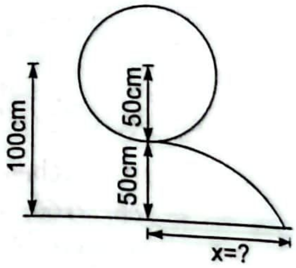
b. How would you verify Hooke's law experimentally? [3]

c. A spherical ball contracts in volume by 0.1%, when subjected to a normal uniform pressure of 100 atmospheres. Calculate the bulk modulus of the material of the ball. [3]

Ans: 1.01×10^{10} Nm^{-2}

OR

A stone of mass 500 gm is attached to a string of length 50 cm which will break if the tension in it exceeds 20 N. The stone is whirled in a vertical circle, the axis of rotation being at a height of 100 cm above the ground as shown in fig.



a. In what position is this break most likely to occur and at what angular speed? [3]

b. Where will the stone hit the ground? [2]

c. What would be the centripetal acceleration and centripetal force? [3]

Ans: (a) 7.75 rad/sec. (b) 1.24 m (c) 30 m/sec, 15N

22. a. Draw a plot showing the variation of binding energy per nucleon with mass no. 'A'.

b. Write two important conclusions which you can draw from this plot. [2]

c. Explain with the help of this plot, the released in energy in the processes of nuclear fusion and fission.[2]

d. A nucleus of uranium -238 disintegrates according to the reaction. [3]

$${}_{92}U^{238} \longrightarrow {}_{90}Th^{234} + {}_{2}He^{4}$$

Calculate the K.E. of α-particles, the nucleus being at rest before disintegration.

[Mass of ${}_{92}U^{238}$ = 238.12492 U
Mass of ${}_{90}Th^{234}$ = 234.1165 U
Mass of ${}_{2}He^{4}$ = 4.00387 U, 1 U = 931 MeV]

Ans: 4.16 MeV

■